ICE ENTERTAINMENT

MIS774 Assessment 1 - Case Study

Company	ICE Entertainment
Headquarters	Melbourne, Australia
Founded	1999
Employees	Approximately 500
Revenue	USD $500+ million
Industry	Entertainment Retail — Music, Films, Audio Books
Ownership	Privately held
Stores	500 locations across 10 countries
Customers	5,000 registered
Data Period	January 2022 – March 2026 (Data Extract)

1. ICE Entertainment - Overview

ICE Entertainment is a privately held entertainment retailer headquartered in Melbourne, Australia. Founded in 1999, the company has grown from a single Australian music store into one of the most geographically diverse digital entertainment retailers in the world, with a catalogue spanning music, films, and audiobooks. ICE Entertainment operates 500 store locations across ten countries — the United States, Germany, France, the United Kingdom, Spain, Australia, Italy, South Africa, China, and India — comprising a mix of Full Outlets, Mini Outlets, and Distribution Centres.

The business model combines individual content purchases, once-off streaming, and subscription packages, making ICE Entertainment one of the few entertainment retailers to successfully compete across all three modes of digital content delivery. The company services customers in multiple currencies — Australian Dollars (AUD), US Dollars (USD), Euros (EUR), British Pounds (GBP), Indian Rupees (INR), South African Rand (ZAR), and Chinese Yuan (CNY) — with all management reporting standardised to US Dollars.

> *"The objective is to provide a global truth — a single source of data for all managerial reports."*
> — Dhil Rodnut, CEO, ICE Entertainment

2. Products and Services

ICE Entertainment offers three categories of digital content as their main products: Music, Films, and Audio Books. Products are offered to customers through three distinct acquisition modes:

Individual Purchase

Customers purchase a single product at its listed unit price. The product is downloaded and owned permanently. Prices range from $0.69 for individual music tracks to $12.99 for premium audio books and films.

Once-Off Streaming

Customers can stream any product once for one-tenth of its purchase price. A film priced at $5.00 can be streamed for $0.50. Streaming is delivered via Geo Broadcasting Ltd., a central infrastructure provider that invoices ICE Entertainment monthly based on actual usage volume. A reliable internet connection is required.

Subscription Packages

ICE Entertainment offers 24 subscription packages across five package types: General Mix, Specific Mix, Single Type, Together (multi-user), and Retention (win-back) packages. Popular packages include:

Package Name	Description	Monthly Price (USD)
Basic	Download any 50 items per month, stream anything unlimited	$19.99
Premium	Download any 100 items per month, stream anything unlimited	$24.99
Ultra	Download any 200 items per month, stream anything unlimited	$29.99
Music 123	Unlimited download or stream any music	$5.99
Films 123	Unlimited download or stream any films	$12.99
Book 123	Unlimited download or stream any audio books	$6.99
Together Premium	Download any 100 items each for two people	$34.99
New Subscriber	3-month introductory package — download or stream anything	$19.99

Payment methods for subscriptions include monthly direct debit, quarterly in advance, and annual in advance.

3. Business Operations

3.1 Store Network

ICE Entertainment's 500 store locations are classified into three types: Full Outlets, Mini Outlets, and Distribution Centres. The store network spans all ten operating countries and is organised into two geographic divisions — AAA (America, Asia and Australia) and EMEA (Europe, Middle East and Africa) — with individual regions assigned within each division (e.g. Australia, Central US, Germany, United Kingdom, South Africa).

In Australia, stores are further classified by state, supporting the company's three-tier market segmentation model: Victoria, Rest of Australia, and International. This market classification is a critical dimension for management reporting and is subject to periodic redefinition as the business evolves.

3.2 Procurement

ICE Entertainment purchases content in bulk from suppliers. For example, the company may negotiate a deal to acquire 10,000 songs from a record label for a fixed total cost, regardless of individual title. The unit cost of any product in the system is derived from these bulk purchase orders and is captured at the time the product is loaded into the ERP system. The ERP automatically manages most reorder decisions, though senior managers occasionally intervene based on commercial judgement.

For online streaming, ICE Entertainment does not own the streaming infrastructure. Instead, it contracts with Geo Broadcasting Ltd. and is invoiced monthly based on actual streaming volumes. This creates a variable cost component that sits outside the standard product cost model and represents an important consideration when calculating true product profitability.

3.3 Delivery Channels

Content is delivered to customers through four main channels: Internet (direct download and streaming), mobile phone, cable TV, and post (physical media for legacy customers). Customers nominate up to two preferred contact channels for CRM communications — choices include Email, Post, Telephone, and RSS. Channel preferences are stored against each customer record and are a key input for the proposed loyalty program.

4. Strategic Context and Business Challenges

ICE Entertainment has delivered strong revenue growth over the past five years, with FY2023 revenue exceeding USD $500 million. However, the pace of growth has outstripped the company's ability to manage and interpret its own data. The CEO, Dhil Rodnut, has acknowledged that despite the volume and richness of data captured across the Sales and ERP systems, management decisions are still largely driven by intuition rather than evidence.

> *"We have data everywhere. The problem is we cannot see it all in one place at the same time. By the time we compile a global sales report, the market has already moved."*
> — Pearce Noj, Sales Manager, ICE Entertainment

The board has approved a comprehensive enterprise-wide analytical solution (dimensional model) initiative to address this. The initiative has three strategic priorities, each tied to a specific business domain:

Priority 1 — Global Sales Visibility (Sales Domain)

ICE Entertainment cannot currently produce a consolidated global sales and profit report in real time. Revenue data sits in the Sales system (Informix) in local currencies. Cost data sits in the ERP (Oracle). Neither system speaks directly to the other, and currency conversion is performed manually and inconsistently across regions. The board wants a single, trusted view of worldwide revenue, cost, and margin — by product, by store, by customer, and by time period — with all figures expressed in USD.

Priority 2 — Supplier Performance and Inventory Intelligence (Purchasing Domain)

The purchasing team currently has no structured way to evaluate whether suppliers are delivering value. Products are purchased in bulk with limited visibility into which titles within a bulk order are actually generating revenue. Milton, the Head of Purchasing, believes the company is over-investing in certain content categories and under-investing in others, but cannot prove it without better data. The dimensional model should enable analysis of product performance at the category and type level to inform future procurement decisions.

Priority 3 — Customer Loyalty Program (CRM Domain)

ICE Entertainment is planning to launch its first customer loyalty program within the next 18 months. However, the CRM team lacks the foundational data infrastructure to design or operate the program effectively. Before a loyalty program can be built, the business needs to understand: who its most valuable customers are, what they buy, how often, through which channel, and whether their behaviour changes when they move house or change their subscription. The dimensional model will provide the analytical foundation for the loyalty program design.

Key Strategic Tension:
ICE Entertainment faces a classic data integration challenge. Two source systems — the Sales System (Informix) and the ERP System (Oracle) — capture complementary but disconnected slices of the business. The Sales system knows what was sold, to whom, when, and where. The ERP system knows what products cost and what was in stock. Neither system on its own can answer the question that matters most to the board: how profitable are we, really, by product, customer, and market?

5. The Insights Required for Decision-Making

Decisions about the future direction of the business are made by Rod (CEO), with input from Pearce (Sales Manager) and Milton (Head of Purchasing). Although both Pearce and Milton have deep experience and strong commercial instincts, Rod wants decisions grounded in data. The dimensional model must be designed to answer the following five business questions.

Question 1 — Who are the key customers?

The dimensional model must provide unit sales, dollar sales, and margin by customer for various time periods including year, quarter, and month. It must also show the postcode in which the most profitable customers live. Note that customers occasionally move house — it is important that the dimensional model records the location at the time of each sale, not the customer's current address. Customer reports must support aggregation into three age groups: 30 years and under, 31 to 50 years, and over 50 years. All monetary figures must be reported in USD, with local currency transactions converted using the applicable monthly exchange rate.

Calculation definitions:
Dollar Sales = qty × price (in original transaction currency, converted to USD)
Cost = unit_cost × qty (using the cost of the product on the most recent purchase order prior to the sale date)
Margin = Dollar Sales (USD) − Cost (USD)

Question 2 — Which products are the most profitable?

The dimensional model must provide unit sales, dollar sales, and margin by product for various time periods including year, season, and month. Products should be navigable by category (Music, Films, Audio Books), by type (e.g. Classical Music, Drama Films, Fiction Books), and by individual product title. Seasonal analysis is important for identifying content purchasing patterns — seasons are defined as: Summer (December–February), Autumn (March–May), Winter (June–August), and Spring (September–November).

Question 3 — Which store locations are the most profitable?

The dimensional model must provide unit sales, dollar sales, and margin for each store location for each month. Store performance should be navigable through a geographic hierarchy from the highest level of aggregation down to the individual store: Division → Region → Country → State (for Australian stores) → City → Store. Store type (Full Outlet, Mini Outlet, Distribution Centre) should also be available as a filter.

Question 4 — Which time periods are the most profitable?

The dimensional model must support time-based analysis of unit sales, dollar sales, and margin across the following granularities: daily (with particular interest in Sundays and public holidays), weekly, monthly, quarterly, and yearly. ICE Entertainment operates on a fiscal year that runs from 1 September to 31 August. Fiscal year, fiscal quarter, and fiscal period reporting are required in addition to calendar-based time analysis.

Fiscal Year Definition:
FY2024 = 1 September 2023 to 31 August 2024
FY2025 = 1 September 2024 to 31 August 2025
FY2026 = 1 September 2025 to 31 August 2026

Question 5 — Which market is the most profitable?

The dimensional model must provide unit sales and dollar sales for each market for each month. ICE Entertainment currently segments its business into three markets based on customer location: Victoria (customers and stores in the state of Victoria, Australia), Rest of Australia (all other Australian customers and stores), and International (all customers and stores outside Australia). These market definitions may change once every three to five years as the business expands into new territories. The design must accommodate this without requiring historical data to be reprocessed.

6. Source Systems

Data for the ICE Entertainment dimensional model will be sourced from two existing operational systems. These systems were built independently, use different database platforms, and store data in incompatible formats. Integrating them is one of the primary technical challenges of this project.

6.1 The Sales System (Informix)

The Sales system manages all customer-facing transactions including individual product sales, subscriptions, and customer master data. Key tables are shown below with sample data.

CUSTOMER (sample data, incomplete)

customer_number	name	gender	address1	zipcode	country	Market
10000	Jessica Davis	M	96 StreetName224, Sydney, Australia	9279	AU	Rest of Australia
10001	Mary Clothilde	F	368 StreetName868, London, England	5333	GB	International
10002	Ethan Miller	M	285 StreetName465, Glasgow, Scotland	6977	GB	International
10003	Wei Thomas	M	33 StreetName825, Paris, France	7572	FR	International

SALES ORDER HEADER (sample data, incomplete)

order_id	order_date	customer_number	store_number	currency
1001	2024-01-06	14515	1451	AUD
1003	2023-06-20	14890	1319	EUR
1006	2024-12-23	12775	1091	CNY
1010	2022-06-27	13529	1382	EUR

SALES ORDER DETAIL (sample data, incomplete)

order_id	line_no	product_code	qty	price	unit_cost
1001	1	100408	2	4.18	1.52
1001	2	101912	1	0.88	0.54
1001	3	101459	2	5.87	2.91
1001	4	100754	4	0.86	0.44
1003	1	100854	1	0.71	0.28
1003	2	102790	2	5.97	3.69

STORE (sample data, incomplete)

store_number	store_name	store_type	address1	zipcode	country
1010	Dusseldorf5	Distribution Centre	Street Name 203	N7710R	DE
1011	Hyderabad8	Mini Outlet	Street Name 748	P9401N	IN
1012	Guangzhou20	Mini Outlet	Street Name 743	L3388F	CN
1013	Birmingham12	Full Outlet	Street Name 200	P7646E	GB

6.2 The ERP System (Oracle)

The ERP system manages all product, inventory, and financial data. Key tables are shown below with sample data.

PRODUCT (sample data, incomplete)

product_code	product_type	product_category	format	unit_price	unit_cost	status
100000	Urban Music	Music	ALAC	0.90	0.46	AC
100003	Rumba Music	Music	WAV	5.03	2.19	AC
100408	Indie Music	Music	FLAC	4.18	1.52	AC
101912	Horror Films	Films	MPEG-4	0.88	0.54	AC
101459	Fiction Books	Audio Books	MP3	5.87	2.91	AC
100854	Classical Music	Music	WAV	0.71	0.28	AC

Note: unit_price is the price of one unit of the product in its original currency. unit_cost is the cost allocated to the product from the most recent bulk purchase order prior to the sale date. Status codes: AC = Active, IN = Inactive, QC = Quality Control, WD = Withdrawn. Only Active products appear in sales transactions.

CURRENCY RATE (sample data, incomplete)

effective_date	currency_code	currency_rate	created	last_updated
2022-01-01	AUD	0.6419	2021-12-26	2021-12-31
2022-01-01	EUR	1.0800	2021-12-26	2021-12-31
2022-01-01	GBP	1.2700	2021-12-26	2021-12-31
2022-01-01	CNY	0.1380	2021-12-26	2021-12-31
2022-02-01	AUD	0.6464	2022-01-25	2022-01-31
2022-02-01	EUR	1.0452	2022-01-25	2022-01-31

Note: currency_rate is the conversion factor to multiply by to convert a local currency amount to USD. Rates are updated monthly. US Dollar (USD) is the base reporting currency and does not appear in this table — USD transactions require no conversion.